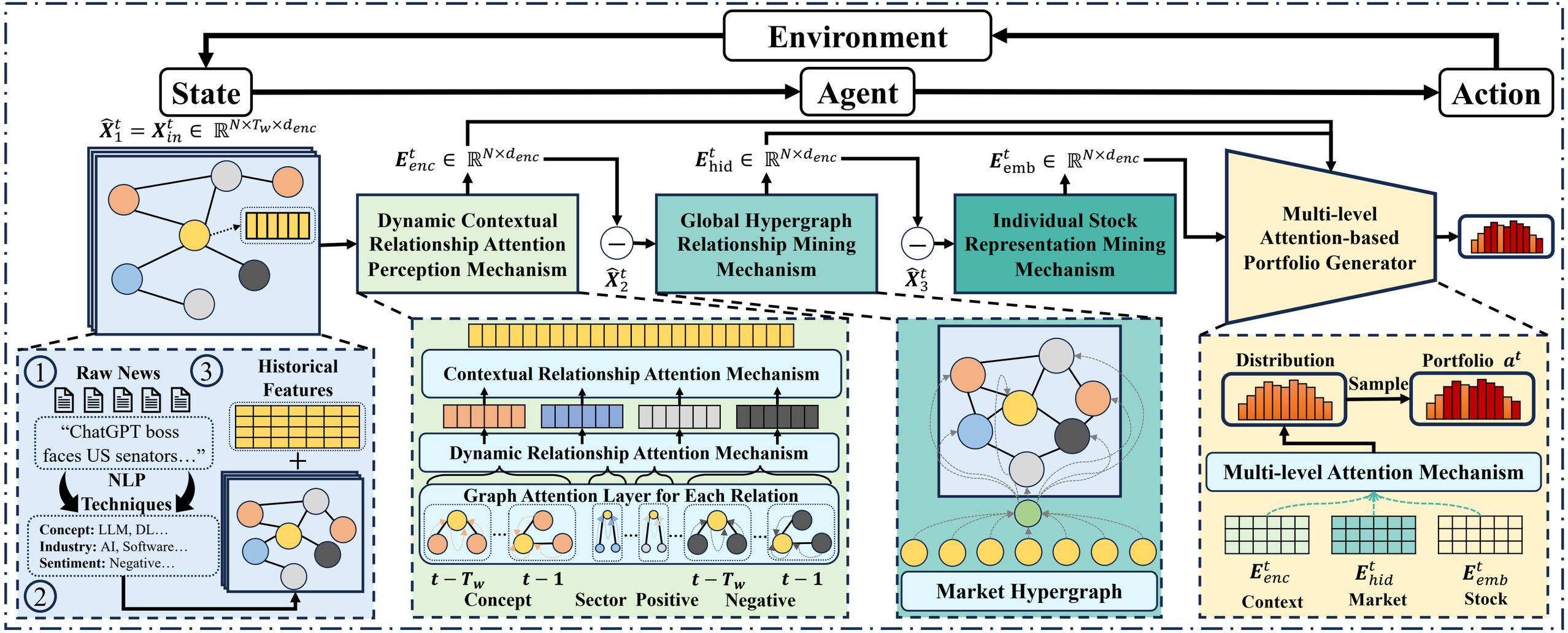
Environment
State
Agent
Action
$\hat{X}_1^t = X_{in}^t \in \mathbb{R}^{N \times T_w \times d_{enc}}$
$E_{enc}^t \in \mathbb{R}^{N \times d_{enc}}$
$E_{hid}^t \in \mathbb{R}^{N \times d_{enc}}$
$E_{emb}^t \in \mathbb{R}^{N \times d_{enc}}$
Dynamic Contextual Relationship Attention Perception Mechanism
$\hat{X}_2^t$
Global Hypergraph Relationship Mining Mechanism
$\hat{X}_3^t$
Individual Stock Representation Mining Mechanism
Multi-level Attention-based Portfolio Generator
1 Raw News
3 Historical Features
"ChatGPT boss faces US senators…"
NLP Techniques
Concept: LLM, DL…
Industry: AI, Software…
Sentiment: Negative…
2
Contextual Relationship Attention Mechanism
Dynamic Relationship Attention Mechanism
Graph Attention Layer for Each Relation
$t - T_w$ Concept
$t - 1$ Sector
$t - T_w$ Positive
$t - 1$ Negative
Market Hypergraph
Distribution
Sample
Portfolio a^t
Multi-level Attention Mechanism
E_{enc}^t Context
E_{hid}^t Market
E_{emb}^t Stock